

GNN Model With Robust Finite-Time Convergence for Time-Varying Systems of Linear Equations

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Abstract—Dynamic neural networks are considered as an effective method in the field of scientific computing, among which gradient neural networks (GNNs) are an efficient method for solving static problems. However, when solving dynamic problems, the current GNNs are often subject to lagging errors. In this article, we develop a finite-time convergent GNN (FTCGNN) model for solving static and time-varying systems of linear equations. Different from zeroing neural networks (ZNNs) dedicated to time-varying problem solving, the FTCGNN model has finite-time convergence regardless of the existence of time-varying noises. Simulation results show that the FTCGNN model is effective, among which the comparisons with existing GNNs and ZNNs validate the advantages of the FTCGNN model.

Index Terms—Finite-time convergence, gradient neural network (GNN), neurodynamics, systems of linear equations (SLEs), time-varying problems.

I. INTRODUCTION

DYNAMIC neural networks (DNNs) are modeled as ordinary differential equations. The benefit of DNNs lies in its convergence guarantee without off-line training based on large amount of data. Such neural networks are often with state feedbacks and are called recurrent neural networks (RNNs). The most classical RNN should be the Hopfield neural network, which is named after its inventor Hopfield. Hopfield and Tank [1] showed that DNNs can efficiently solve the traveling-salesman problem that is NP-complete. The early work of Hopfield and Tank provides a new perspective for solving scientific computing problems by leveraging the advantages of DNNs. Technically, the Hopfield neural network relies on the minimization of an

energy function, by using its gradient information. Following this idea, many DNNs based on optimization theory have been proposed. In particular, Xia and Wang [2] proposed a general methodology for designing globally convergent RNNs for solving nonlinear constrained optimization problems, of which the global asymptotic convergence of the state variable to the theoretical solution is theoretically guaranteed by using Lyapunov stability theory [3]. DNNs have already been adopted for optimization problems in robotics [4], control [5], image processing [6], speech enhancement [7], task assignment [8], feature selection [9], and distributed chiller loading [10].

While the classical Hopfield neural network targeting at optimization problems, many novel DNN models have been proposed to solve other types of scientific computing problems. Cichocki and Unbehauen [11] first developed RNNs to handle static systems of linear equations (SLEs). Wang and Li [12] proposed a linear-function-activated RNN to address linear equations, and the network has global asymptotic convergence. It should be noted that the RNN developed in [12] is a special case of the corresponding one proposed in [11]. Essentially, the RNN proposed in [12] is a gradient neural network (GNN) as its design leverages the energy function's gradient information. Zhang et al. [13] theoretically analyzed the convergence behavior of the GNN in [12] with different activation functions. Xia et al. [14] proposed globally convergent RNNs to address systems of linear inequalities and equations by converting the problem into an optimization problem. Zhang et al. [15] proposed an RNN to handle dynamic Sylvester equations, and the RNN is recognized as the zeroing neural network (ZNN) in the existing literature. The solution error of the ZNN has exponential convergence. Unlike the Hopfield neural network or GNN, the RNN proposed in [15] is described by an implicit dynamic equation and its design is not based on a non-negative energy function. Instead, the RNN proposed in [15] relies on an error function that can be matrix- or vector-valued and neural dynamics is then designed for exponential error convergence. In terms of model complexity, the ZNN is more complex than the Hopfield neural network, because the ZNN fully employs the time-derivative information of all the matrices, vectors, and scalars involved in the computing problems. The RNN proposed in [15] has been adopted to solve many scientific computing problems, including dynamic matrix inversion [16], dynamic generalized inversion [17], dynamic linear equation and inequality systems [18], dynamic linear equations subject to bound constraints [19], and time-varying linear equations [20].

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Zhang et al.'s RNN [15] has been improved by mainly two approaches. First, from the perspective of control theory, an integration term is introduced to the original design formula, by which the robustness of the neural network to noises is improved [21], [22], [23], [24]. Second, an interesting activation function is developed by Li et al. [25] such that the RNN [15] converges in finite time when handling dynamic Sylvester equations, i.e., the state variable of the RNN can track the theoretical solution after certain time. Note that, although finite-time stability has been widely investigated [26], [27], [28], [29], [30], [31], most of the results focus on the design methods of a finite-time controller for control systems, and the extension of such results to the design of neural networks of finite-time convergence when handling scientific computing problems, such as the dynamic SLEs, is not trivial. In the design of RNNs for solving scientific computing problems, apart from the stability of the RNN models, we must ensure that the state variable converges to the corresponding theoretical solution. Following Li et al.'s work [25], Xiao and Lu [32], Xiao et al. [33], Dai et al. [34], and Lv et al. [35] proposed finite-time convergent RNNs for solving various time-varying scientific computing problems. Xiao et al. [36] also combined the above two improvements, and further designed a robust finite-time convergent RNN for solving dynamic linear equations. It should be noted that both improvements make the model of the ZNN become more complex.

It is widely recognized that GNNs have intrinsic disadvantages when they are adopted to solving time-varying problems [37], [38], [39], [40]. For example, Xiao and Zhang [40] showed that the traditional GNN is not favorable for handling dynamic linear inequalities, and they argued that this is due to the fact that GNN ignores time-derivative information in the problem. In [41], for calculating the inverse of a dynamic matrix, an adaptive coefficient, including time-derivative information, is introduced into the original GNN. Owing to the adaptive coefficient, the improved GNN has asymptotic convergence for time-varying dynamic matrix inversion, which is still not competent compared with the ZNN of exponential convergence.

In this article, we provide a counter-intuition result for the design of GNN. Specifically, we show that GNN can solve time-varying SLEs in finite-time without knowing the exact derivatives or partial derivatives about the SLEs. Evidently, this result is more promising than the result of [41] in which only asymptotic convergence of GNN for solving a time-varying problem is achieved. Moreover, we also show that the the proposed GNN has satisfactory robustness for the case with noises. The major contributions in our work include the following ones.

- 1) A unified GNN model is first developed for solving both static and dynamic SLEs with proven finite-time convergence.
- 2) The proposed GNN model is of simpler structure compared with existing ZNN [24] dedicated to solving time-varying SLEs under noises; unlike ZNNs, such as [36], our GNN model does not require knowing explicit formulas or exact values

of time-derivatives of variables involved in the SLEs.

- 3) Our GNN model has finite-time convergence when solving dynamic SLEs under additive bounded time-varying noises by setting the design parameter.

We describe the considered problem and review the existing GNN models for solving SLEs in Section II. Then, the design and theoretical analysis of our GNN model are shown in Section III. Simulation results and performance comparison with existing ZNN and GNN models are shown in Section IV, followed by conclusions and future works given in Section V.

II. PROBLEM DESCRIPTION AND EXISTING GNN MODELS

In this section, we introduce the investigated problem and revisit some existing GNN models for solving SLEs.

A. Problem Description

An SLE can be described as follows:

$$\mathbf{A}\mathbf{x} - \mathbf{b} = \mathbf{0} \in \mathbb{R}^m \quad (1)$$

in which $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$ are a full-rank matrix with $m \geq n$ and a vector, respectively. The vector $\mathbf{x} \in \mathbb{R}^n$ is the unknown variable. Let $\mathbf{x}^*$ denote a theoretical solution of (1). We assume that the theoretical solution exists. When $\mathbf{A}$ and $\mathbf{b}$ are independent of time t , the SLE is static; otherwise, it is called a time-varying or dynamic SLE.

B. Existing GNN for SLEs

In this section, we review the existing GNNs for solving SLE (1).

Basically, the design of GNNs relies on a non-negative energy function. For SLE (1), the corresponding energy function is given as [13]

$$h(\mathbf{x}) = \frac{\|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2}{2}.$$

Evidently, if the energy function reduces to zero, $\mathbf{x}$ is a solution to SLE (1). To this end, we make $\mathbf{x}$ moves along the opposite direction of the gradient of $h(\mathbf{x})$, which is calculated as

$$-\frac{\partial h(\mathbf{x})}{\partial \mathbf{x}} = -\mathbf{A}^T(\mathbf{A}\mathbf{x} - \mathbf{b}).$$

It is known that the convergence of DNNs could be improved by using activation functions. Let $\phi(\cdot) : \mathbb{R}^m \rightarrow \mathbb{R}^m$ denote a monotonically increasing odd function. Then, a GNN model with an activation function is obtained as follows [11], [12], [13]:

$$\dot{\mathbf{x}} = -\gamma \mathbf{A}^T \phi(\mathbf{A}\mathbf{x} - \mathbf{b}) \quad (2)$$

where $\dot{\mathbf{x}} = d\mathbf{x}/dt$ and $\gamma > 0 \in \mathbb{R}$ is used to control the convergence of the state variable $\mathbf{x}$ of the GNN. It is proven that the state variable $\mathbf{x}$ of the GNN (2) exponentially converges to theoretical solution $\mathbf{x}^*$ of the corresponding system of linear equation (1) [13]. There are several widely adopted activation functions that can be used for DNNs, such as the following ones [42], [43], [44].

1) Linear activation function

$$\phi(x) = x. \quad (3)$$

2) Sinh activation function

$$\phi(x) = \frac{\exp(x) - \exp(-x)}{2}. \quad (4)$$

3) Bipolar-sigmoid activation function

$$\phi(x) = \frac{(1 + \exp(-\omega))(1 - \exp(-\omega x))}{(1 - \exp(-\omega))(1 + \exp(-\omega x))}, \text{ with } \omega \geq 2. \quad (5)$$

4) Powersum activation function

$$\phi(x) = \sum_{i=1}^N x^{2i-1}, \text{ with } N \text{ being a positive integer.} \quad (6)$$

5) Sign-bi-power activation function

$$\phi(x) = \text{sgn}^r(x) + \text{sgn}^{1/r}(x) \quad (7)$$

where $r \in (0, 1)$ and $\text{sgn}^r(x)$ is defined as

$$\text{sgn}^r(x) = \begin{cases} |x|^r, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -|x|^r, & \text{if } x < 0. \end{cases}$$

If the linear activation function is adopted, (2) reduces to the RNN proposed in [12]. If the activation functions 2)–5) are adopted, the GNN (2) achieves better convergence. In particular, if the function (7) is used the GNN (2) achieves finite-time convergence, and the corresponding finite-time convergent GNN (FTCGNN) model is given as follows [44]:

$$\dot{\mathbf{x}} = -\gamma A^T (\text{sgn}^r(A\mathbf{x} - \mathbf{b}) + \text{sgn}^{1/r}(A\mathbf{x} - \mathbf{b})) \quad (8)$$

where $r \in (0, 1)$ and $\gamma > 0 \in \mathbb{R}$. We say that $\mathbf{x}$ converges to the theoretical solution $\mathbf{x}^*$ asymptotically if $\lim_{t \rightarrow +\infty} \mathbf{x}(t) = \mathbf{x}^*$. We further say that $\mathbf{x}$ converges to the theoretical solution $\mathbf{x}^*$ in finite time if $\mathbf{x}(t) = \mathbf{x}^*$ for any $t > T$ for a certain $T > 0 \in \mathbb{R}$.

In terms of the FTCGNN model (8) activated by (7), we have the following remark.

Remark 1: By comparing the activation functions of RNNs with finite-time convergence in the existing literature, such as FTCGNN model (8) activated by (7), we find that the activation function similar to the sign function is important in realizing the finite-time convergence of the GNN. The sign function is

$$\text{sign}(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -1 & \text{if } x < 0. \end{cases}$$

In fact, from the existing literature about finite-time control and finite-time convergent neural networks, such as [34], we can find that the sign function or the sign-like function $\text{sgn}^r(x)$ are often adopted. In [34], some sign-function-based activation functions are adopted in the design of finite-time convergent RNNs. The problem of the sign function lies in its discontinuity, which causes chattering in the numerical implementation

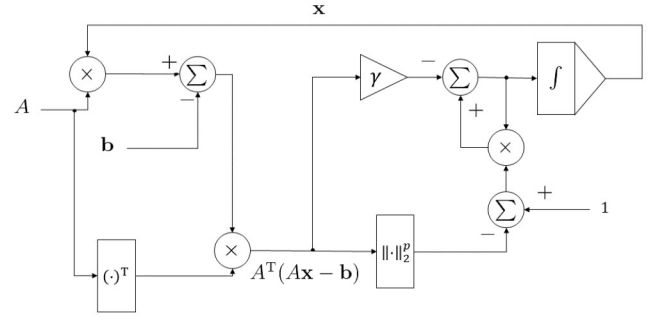


Fig. 1. Block diagram for the hardware implementation of FTCGNN model (9).

of the corresponding models and this phenomenon is widely recognized in the field of sliding mode control [45]. In terms of (7) adopted in FTCGNN (8), although it is continuous, its implementation via circuits is more complicated than the sign function.

III. NEW FTCGNN FOR SLEs

This section provides a new FTCGNN without using sign or sign-like activation functions for solving SLE (1). The theoretical analysis for the model is also provided.

A. Model Description of New FTCGNN

The model of the new FTCGNN is given as follows:

$$\|A^T(A\mathbf{x} - \mathbf{b})\|_2^p \dot{\mathbf{x}} = -\gamma A^T(A\mathbf{x} - \mathbf{b}) \quad (9)$$

in which $\|\cdot\|_2$ is the 2-norm, γ is used to scale the convergence, and $p \in (0, 2)$. Evidently, the new FTCGNN is an RNN owing to the feedback connection of state variable $\mathbf{x}$. The proposed FTCGNN model can be realized by using analog circuits with $\dot{\mathbf{x}} = (1 - \|A^T(A\mathbf{x} - \mathbf{b})\|_2^p) \dot{\mathbf{x}} - \gamma A^T(A\mathbf{x} - \mathbf{b})$. Fig. 1 shows the block diagram regarding the hardware implementation of FTCGNN (9). Intuitively, the proposed FTCGNN model (9) leverages $\|A^T(A\mathbf{x} - \mathbf{b})\|_2^p$ to provide an adaptive adjustment for the evolution of $\mathbf{x}$. Owing to the adaptive feature, the proposed FTCGNN model (9) achieves better performance. This coincides with the theoretical analysis on its convergence and robustness.

Remark 2: The GNN model proposed in [7] is for solving a class of nonlinear optimization problems subject to various constraints, which is formulated as

$$\begin{aligned} \epsilon \dot{x} &= -x + P_X(x - \nabla L_a(x, y, z)) \\ \epsilon \dot{y} &= -y + (y + g(x))^+ \\ \epsilon \dot{z} &= h(x) \end{aligned} \quad (10)$$

where the definitions of the symbols can be found in [7]. Obviously, our FTCGNN model (9) is totally different from the GNN model proposed in [7], i.e., (10). The GNN proposed in [9] is formulated as

$$\dot{\mathbf{x}} = -\gamma A^T(A\mathbf{x} - \mathbf{b}). \quad (11)$$

Obviously, compared with the GNN proposed in [9], an adaptive gain $\|A^T(A\mathbf{x} - \mathbf{b})\|_2^p$ is included in our FTCGNN

model (9). The new FTCGNN (9) does not employ sign or sign-like activation functions. The design of the new FTCGNN model (9) is not trivial, and it is different from the existing work [44] which accelerates the convergence of the traditional GNN (2) by designing a new activation function for the classical GNN (2). It can be found that the new FTCGNN model (9) cannot be readily developed from the classical GNN (2) via choosing monotonically increasing odd activation functions. The FTCGNN model (9) has lower complexity compared with (8) proposed in [44]. Owing to the new design, the theoretical analysis on FTCGNN model (9) cannot be trivially generalized from [44].

B. Convergence of New FTCGNN for Static SLEs

In the following, the convergence analysis on FTCGNN (9) is shown for solving static SLEs.

Theorem 1: If $p \in (0, 2)$, $\gamma > 0$, and the minimum eigenvalue of $A^T A$ is larger than 0, then the state variable $\mathbf{x}$ of FTCGNN model (9) globally converges to a solution $\mathbf{x}^*$ of static SLE (1) in finite time, and the upper bound of the convergence time is $\sqrt{2}^{2-p} \|\tilde{\mathbf{x}}(0)\|_2^p / (\lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \gamma p)$.

Proof: Let $\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{x}^*$ denote the difference between the state variable $\mathbf{x}$ of FTCGNN model (9) and a solution $\mathbf{x}^*$ of SLE (1). Consider

$$V = \frac{\|\mathbf{x} - \mathbf{x}^*\|_2^2}{2} = \frac{\|\tilde{\mathbf{x}}\|_2^2}{2}.$$

Evidently, V is non-negative and radially unbounded since $V \geq 0 \forall \tilde{\mathbf{x}}$ and $V \rightarrow +\infty$ if $\|\tilde{\mathbf{x}}\|_2 \rightarrow +\infty$. Because $\mathbf{x}^*$ is a solution of SLE (1), we have $A\mathbf{x}^* = \mathbf{b}$. With $\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{x}^*$ and $A\mathbf{x}^* = \mathbf{b}$, the new FTCGNN model (9) is rewritten in the following form with respect to $\tilde{\mathbf{x}}$:

$$\|A^T A \tilde{\mathbf{x}}\|_2^p \dot{\tilde{\mathbf{x}}} = -\gamma A^T A \tilde{\mathbf{x}}. \quad (12)$$

If $\|A^T A \tilde{\mathbf{x}}\|_2^p = 0$, it means $A^T A \tilde{\mathbf{x}} = \mathbf{0}$, which means that $\tilde{\mathbf{x}} = \mathbf{0}$ in view of the condition that $\lambda_{\min}(A^T A) > 0$. It follows that $V = 0$. If $\|A^T A \tilde{\mathbf{x}}\|_2^p \neq 0$, the time derivative of V for FTCGNN (9), i.e., (12), is obtained

$$\begin{aligned} \dot{V} &= \tilde{\mathbf{x}}^T \dot{\tilde{\mathbf{x}}} \\ &= -\gamma \tilde{\mathbf{x}}^T A^T A \tilde{\mathbf{x}} \|A^T A \tilde{\mathbf{x}}\|_2^{-p}. \end{aligned}$$

Let $\lambda_{\min}(A^T A)$ denote the minimum eigenvalue of $A^T A$, which is assumed to be larger than 0. Let $\lambda_{\max}(A^T A)$ denote the maximum eigenvalue of $A^T A$. Recall that the eigenvalues of $A^T A$ are identical to AA^T . Then, we further have

$$\begin{aligned} \dot{V} &\leq -\gamma \lambda_{\min}(A^T A) \|\tilde{\mathbf{x}}\|_2^2 \lambda_{\max}^{-p}(A^T A) \|\tilde{\mathbf{x}}\|_2^{-p} \\ &= -\lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \gamma \|\tilde{\mathbf{x}}\|_2^{2-p} \\ &\leq -\lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \gamma V^{\frac{2-p}{2}}. \end{aligned}$$

Let $\beta = \lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \gamma$. It follows that:

$$V^{\frac{p-2}{2}} dV \leq -\beta dt.$$

Following the steps in [25], the integration over both sides from $t = 0$ to $T > 0$ yields:

$$\int_{V(0)}^{V(T)} V^{\frac{p-2}{2}} dV \leq \int_0^T -\beta dt$$

and we further have

$$\frac{2}{p} \left(V^{\frac{p}{2}}(T) - V^{\frac{p}{2}}(0) \right) \leq -\beta T.$$

Thus, we have

$$V^{\frac{p}{2}}(T) \leq V^{\frac{p}{2}}(0) - \frac{\lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \gamma p T}{2}. \quad (13)$$

Let $V^{(p/2)}(0) - \lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \gamma p T^*/2 = 0$, and we have

$$\begin{aligned} T &= \frac{2V^{\frac{p}{2}}(0)}{\lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \gamma p} \\ &= \frac{\sqrt{2}^{2-p} \|\tilde{\mathbf{x}}(0)\|_2^p}{\lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \gamma p}. \end{aligned}$$

In view of $V \geq 0$ by the definition of V , $p \in (0, 2)$, $\gamma > 0$, $T > 0$, and (13), we further have

$$V(t) = 0 \quad \forall t > \frac{\sqrt{2}^{2-p} \|\tilde{\mathbf{x}}(0)\|_2^p}{\lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \gamma p}.$$

When $V(t) = 0$, we have $\tilde{\mathbf{x}} = \mathbf{0}$. By summarizing the above analysis, $\mathbf{x}$ of FTCGNN (9) globally converges to $\mathbf{x}^*$ of static SLE (1) in finite time, and the upper bound of the convergence time is $\sqrt{2}^{2-p} \|\tilde{\mathbf{x}}(0)\|_2^p / (\lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \gamma p)$. The proof is complete. ■

According to the above analysis, we can adjust the convergence time by setting the parameters p and γ of the proposed FTCGNN (9). In particular, the convergence time is almost inversely proportional to γ . Regarding the condition that $\lambda_{\min}(A^T A) > 0$, by which $A^T A$ is invertible, although in this case we can solve the SLE via $\mathbf{x} = (A^T A)^{-1} A^T \mathbf{b}$, the computational cost is very large owing to the inversion operation when the dimension of matrix A is large [16], [47]. Note that the general computational complexity of the matrix inversion by using numerical algorithms is $O(n^3)$ [48]. Thus, the direct solution via $\mathbf{x} = (A^T A)^{-1} A^T \mathbf{b}$ is not suitable for large-scale problems. Moreover, such a disadvantages would be enlarged when A and $\mathbf{b}$ are time-varying, for which the real-time calculation of $\mathbf{x}$ via $(A^T A)^{-1} A^T \mathbf{b}$ becomes more difficult owing to the time consumption for the inversion operator [16].

C. Convergence of New FTCGNN for Time-Varying SLEs

In the following, we analyze the performance of the FTCGNN (9) for time-varying SLE. For the convenience of presentation, the argument t is omitted somewhere if there is no confusion.

Assumption 1: The theoretical solution $\mathbf{x}^*(t)$ of time-varying SLE (1) satisfies $\|\mathbf{x}^*(t)\|_2 \leq \eta \forall t \in [0, +\infty)$, where $\eta > 0 \in \mathbb{R}$ is known.

Theorem 2: Consider time-varying SLE (1), for which Assumption 1 holds. If $p = 1$, $\gamma > \eta \lambda_{\max}(A^T A) / \lambda_{\min}(A^T A)$, and the minimum eigenvalue of $A^T A$ is larger than 0, then the state variable $\mathbf{x}$ of FTCGNN model (9) can globally track the dynamic theoretical solution $\mathbf{x}^*(t)$ in finite time with the convergence time being less than $\|\tilde{\mathbf{x}}(0)\|_2 / \mu$ and $\mu = \gamma \lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) - \eta$, i.e., $\tilde{\mathbf{x}}(t) = \mathbf{x}(t) - \mathbf{x}^*(t) = 0$ for any $t > \|\tilde{\mathbf{x}}(0)\|_2 / \mu$.

Proof: Consider

$$V = \frac{\|\mathbf{x} - \mathbf{x}^*\|_2^2}{2} = \frac{\|\tilde{\mathbf{x}}\|_2^2}{2}.$$

In view of the proof in Theorem 1, when solving dynamic SLE (1), i.e., $A(t)\mathbf{x}(t) = \mathbf{b}(t)$, if $\|A^T A \tilde{\mathbf{x}}\|_2^p = 0$, it means $A^T A \tilde{\mathbf{x}} = \mathbf{0}$, which indicates $\tilde{\mathbf{x}} = \mathbf{0}$ in view of the condition that $\lambda_{\min}(A^T A) > 0$. It follows that $V = 0$. If $\|A^T A \tilde{\mathbf{x}}\|_2^p \neq 0$, via the proposed FTCGNN model (9), we have

$$\begin{aligned} \dot{V} &= \tilde{\mathbf{x}}^T \dot{\tilde{\mathbf{x}}} \\ &= \tilde{\mathbf{x}}^T \left(-\gamma A^T A \tilde{\mathbf{x}} \|A^T A \tilde{\mathbf{x}}\|_2^{-p} - \dot{\mathbf{x}}^* \right) \\ &\leq -\lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \gamma \|\tilde{\mathbf{x}}\|_2^{2-p} + \|\dot{\mathbf{x}}^*\|_2 \|\tilde{\mathbf{x}}\|_2 \\ &\leq -\lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \gamma \|\tilde{\mathbf{x}}\|_2^{2-p} + \eta \|\tilde{\mathbf{x}}\|_2 \\ &= -\|\tilde{\mathbf{x}}\|_2 \left(\gamma \lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \|\tilde{\mathbf{x}}\|_2^{1-p} - \eta \right). \end{aligned}$$

If $p = 1$, then we have

$$\dot{V} \leq -\|\tilde{\mathbf{x}}\|_2 \left(\gamma \lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) - \eta \right). \quad (14)$$

Let $\mu = \gamma \lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) - \eta$. Based on inequality (14), if $\gamma \lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) > \eta$, i.e., $\gamma > \eta \lambda_{\max}(A^T A) / \lambda_{\min}(A^T A)$, then $\mu > 0$, by which $\dot{V} \leq 0$. Then, from (14) and the definition of V , we further have

$$\dot{V} \leq -\sqrt{2}\mu V^{\frac{1}{2}}.$$

Solving the above inequality yields

$$V^{\frac{1}{2}}(t) \begin{cases} \leq V^{\frac{1}{2}}(0) - \frac{\mu t}{\sqrt{2}}, & \text{if } t < \frac{\sqrt{2}V^{\frac{1}{2}}(0)}{\mu} \\ = 0, & \text{if } t \geq \frac{\sqrt{2}V^{\frac{1}{2}}(0)}{\mu} \end{cases}$$

i.e.,

$$\|\tilde{\mathbf{x}}(t)\|_2 \begin{cases} \leq \|\tilde{\mathbf{x}}(0)\|_2 - \mu t, & \text{if } t < \frac{\|\tilde{\mathbf{x}}(0)\|_2}{\mu} \\ = 0, & \text{if } t \geq \frac{\|\tilde{\mathbf{x}}(0)\|_2}{\mu}. \end{cases}$$

Then, in light of the radially unbounded property of V , it is concluded that the state variable $\mathbf{x}$ of FTCGNN model (9) can globally track theoretical solution $\mathbf{x}^*(t)$ in finite time, and the convergence time is less than $\|\tilde{\mathbf{x}}(0)\|_2 / \mu$ with $\mu = \gamma \lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) - \eta$. The proof is complete. ■

Theorem 2 provides a very interesting result about using GNNs for dynamic SLEs. Currently, in the literature of time-varying problem solving, researchers argued that using GNNs to address dynamic problems is often subject to lagging errors [37], [38], [39], [40]. In Theorem 2, we show that the designed GNN model (9) can handle time-varying SLEs with finite-time convergence. In practice, the varying rate of solutions to SLEs is always limited. Given that there is a proper estimation of the upper bound of $\|\dot{\mathbf{x}}^*(t)\|_2$, our GNN can address the problem by setting a large value of γ in order to ensure that $\mu > 0$. We recall the classical ZNNs for solving SLEs. For example, in [20], a ZNN model for dynamic SLEs is presented as

$$A\dot{\mathbf{x}} = -\dot{A}\mathbf{x} - \gamma\phi(A\mathbf{x} - \mathbf{b}) + \dot{\mathbf{b}} \quad (15)$$

where $\dot{A} = dA/dt$ and $\dot{\mathbf{b}} = d\mathbf{b}/dt$. In [34], a finite-time convergent ZNN (FTCZNN) model for dynamic SLEs is presented as

$$A^T A \dot{\mathbf{x}} = -A^T \dot{A} \mathbf{x} - \gamma A^T \phi(A\mathbf{x} - \mathbf{b}) + A^T \dot{\mathbf{b}} \quad (16)$$

where the alternative activation functions $\phi(\cdot)$ are given in [34]. Other ZNN models for dynamic SLEs are also proposed in [35], [36], and [46]. The difference or advantages of the proposed FTCGNN model (9) compared with ZNN models (15) and (16), and those in [35], [36], and [46] for solving time-varying SLEs are listed as follows.

- 1) Different from ZNN models (15) and (16), and those in [35], [36], and [46], the proposed FTCGNN model (9) does not need to know the values or explicit formulas of $\dot{A}$ and $\dot{\mathbf{b}}$, and has global finite-time convergence given that a prior knowledge on the upper bound of $\|\dot{\mathbf{x}}^*\|_2$ is available. It should be noted that get the derivatives of variables are sensitive to noises, especially in circuit implementation.
- 2) Compared with the existing FTCZNN models, such as [34], the proposed FTCGNN model (9) does not rely on sign or sign-like activation functions and has a simpler structure.

D. Robustness of New FTCGNN to Noises When Solving Time-Varying SLEs

In [24], to deal with unavoidable noises in hardware implementation, the ZNN model (15) is improved by leveraging control theory via introducing the integration operator, and a noise-tolerant ZNN (NTZNN) model is derived. Let the time-varying noise be denoted by $\sigma(t)$. The following is the NTZNN model for solving time-varying SLEs perturbed by noise $\sigma(t)$ [24]:

$$\dot{\mathbf{x}} = A^+ \left(-\gamma(A\mathbf{x} - \mathbf{b}) - \kappa \int_0^t (A(\tau)\mathbf{x}(\tau) - \mathbf{b}(\tau))d\tau - \dot{A}\mathbf{x} + \dot{\mathbf{b}} \right) + \sigma(t) \quad (17)$$

where A^+ denotes the pseudoinverse of A with γ and κ being positive design parameters.

By following [24], the perturbed model of the proposed FTCGNN (9) for solving dynamic SLE $A(t)\mathbf{x}(t) = \mathbf{b}(t)$ is described as follows:

$$\|A^T(A\mathbf{x} - \mathbf{b})\|_2^p \dot{\mathbf{x}} = -\gamma A^T(A\mathbf{x} - \mathbf{b}) + \|A^T(A\mathbf{x} - \mathbf{b})\|_2^p \sigma(t). \quad (18)$$

Theorem 3: Consider dynamic SLE (1), i.e., $A(t)\mathbf{x}(t) = \mathbf{b}(t)$, of which the theoretical solution $\mathbf{x}^*(t)$ to time-varying SLE (1) satisfies $\|\dot{\mathbf{x}}^*(t)\|_2 \leq \eta$ and the noise satisfies $\|\sigma(t)\|_2 \leq \rho \forall t \in [0, +\infty)$. If $p = 1$, $\gamma > (\eta + \rho) \lambda_{\max}(A^T A) / \lambda_{\min}(A^T A)$, and the minimum eigenvalue of $A^T A$ is larger than 0, then the state variable $\mathbf{x}$ of the noise-perturbed FTCGNN model (18) can globally track $\mathbf{x}^*(t)$ in finite time, and the convergence time is less than $\|\tilde{\mathbf{x}}(0)\|_2 / \mu$ with $\mu = \gamma \lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) - \eta - \rho$, i.e., $\tilde{\mathbf{x}}(t) = \mathbf{x}(t) - \mathbf{x}^*(t) = 0$ for any $t > \|\tilde{\mathbf{x}}(0)\|_2 / \mu$.

Proof: Consider

$$V = \frac{\|\mathbf{x} - \mathbf{x}^*\|_2^2}{2} = \frac{\|\tilde{\mathbf{x}}\|_2^2}{2}.$$

With the proof in Theorem 1, if $\|A^T A \tilde{\mathbf{x}}\|_2^p \neq 0$, with $\|\dot{\mathbf{x}}^*(t)\|_2 \leq \eta$ and $\|\sigma(t)\|_2 \leq \rho \ \forall t \in [0, +\infty)$, for the noise-perturbed FTCGNN model (18), we have

$$\begin{aligned} \dot{V} &= \tilde{\mathbf{x}}^T \dot{\tilde{\mathbf{x}}} \\ &= \tilde{\mathbf{x}}^T \left(-\gamma A^T A \tilde{\mathbf{x}} \|A^T A \tilde{\mathbf{x}}\|_2^{-p} + \sigma - \dot{\mathbf{x}}^* \right) \\ &\leq -\lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \gamma \|\tilde{\mathbf{x}}\|_2^{2-p} + \|\sigma - \dot{\mathbf{x}}^*\|_2 \|\tilde{\mathbf{x}}\|_2 \\ &\leq -\lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \gamma \|\tilde{\mathbf{x}}\|_2^{2-p} + (\|\dot{\mathbf{x}}^*\|_2 + \|\sigma\|_2) \|\tilde{\mathbf{x}}\|_2 \\ &= -\lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \gamma \|\tilde{\mathbf{x}}\|_2^{2-p} + (\eta + \rho) \|\tilde{\mathbf{x}}\|_2 \\ &= -\|\tilde{\mathbf{x}}\|_2 \left(\gamma \lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) \|\tilde{\mathbf{x}}\|_2^{1-p} - \eta - \rho \right). \end{aligned}$$

If $p = 1$, then we further have

$$\dot{V} \leq -\|\tilde{\mathbf{x}}\|_2 \left(\gamma \lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) - \eta - \rho \right). \quad (19)$$

Based on the above inequality, if $\gamma \lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) > (\eta + \rho)$, i.e., $\gamma > (\eta + \rho) \lambda_{\max}(A^T A) / \lambda_{\min}(A^T A)$, then

$$\dot{V} \leq 0.$$

In fact, let

$$\mu = \gamma \lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) - \eta - \rho > 0.$$

Then, from (19) and the definition of V , we further have

$$\dot{V} \leq -\sqrt{2}\mu V^{\frac{1}{2}}.$$

Solving the above inequality yields

$$V^{\frac{1}{2}}(t) \begin{cases} \leq V^{\frac{1}{2}}(0) - \frac{\mu t}{\sqrt{2}}, & \text{if } t < \frac{\sqrt{2}V^{\frac{1}{2}}(0)}{\mu} \\ = 0, & \text{if } t \geq \frac{\sqrt{2}V^{\frac{1}{2}}(0)}{\mu} \end{cases}$$

i.e.,

$$\|\tilde{\mathbf{x}}(t)\|_2 \begin{cases} \leq \|\tilde{\mathbf{x}}(0)\|_2 - \mu t, & \text{if } t < \frac{\|\tilde{\mathbf{x}}(0)\|_2}{\mu} \\ = 0, & \text{if } t \geq \frac{\|\tilde{\mathbf{x}}(0)\|_2}{\mu}. \end{cases}$$

Then, in light of the radially unbounded property of V , it is concluded that $\mathbf{x}$ of the noise-perturbed FTCGNN model (18) can globally track $\mathbf{x}^*(t)$ in finite time, and the convergence time is less than $\|\tilde{\mathbf{x}}(0)\|_2 / \mu$ with $\mu = \gamma \lambda_{\min}(A^T A) \lambda_{\max}^{-p}(A^T A) - \eta - \rho$. The proof is complete. ■

Remark 3: There are two parameters in the proposed FTCGNN model, i.e., p and γ . According to Theorem 1, p should satisfy $0 < p < 2$ to have a good performance for solving static SLEs. According to Theorems 2 and 3, p should be set to 1 to have a good performance for solving time-varying SLEs. In terms of γ , Theorems 1–3 indicate that a larger value of γ leads to faster convergence and better robustness to noises.

Remark 4: Compared with NTZNN model (17), the advantages of FTCGNN model (18) for solving time-varying SLEs include the following aspects.

- 1) Compared with NTZNN model (17), the proposed FTCGNN model (18) does not need the explicit formulas

or values of $\dot{A}$ and $\dot{\mathbf{b}}$. Instead, the proposed FTCGNN model (18) only needs to know an upper bound of $\|\dot{\mathbf{x}}^*(t)\|_2$, which may be estimated in practical applications by using available prior knowledge.

- 2) Compared with NTZNN model (17), the proposed FTCGNN model (18) does not need the integration operator to realize the robustness to noises. By comparing (17) with (18), it can be readily found that the FTCGNN model has a simpler structure.
- 3) Under bounded noises, the NTZNN model (17) cannot achieve asymptotic convergence for solving time-varying SLEs, while the proposed FTCGNN model (18) can achieve finite-time convergence under mild conditions.

Remark 5: The limitation of the proposed method lies in the requirement on knowing the upper bound of $\|\dot{\mathbf{x}}^*(t)\|_2$, i.e., the Euclidian norm of the time derivative of the theoretical solution to the time-varying SLE. According to Theorems 2 and 3, to ensure finite-time convergence of the proposed FTCGNN model, γ should be larger than a value that depends on the upper bound of $\|\dot{\mathbf{x}}^*(t)\|_2$. In practice, for some applications, we can determine the upper bound of $\|\dot{\mathbf{x}}^*(t)\|_2$ based on the physical meaning of $\dot{\mathbf{x}}^*$. For example, for the manipulator application shown in Section IV-E of this article, we have prior knowledge about the maximum joint acceleration $\ddot{\theta}$ for a given physical manipulator, and we can use this prior knowledge to get an upper bound of $\|\dot{\mathbf{x}}^*(t)\|_2$. For the case that such prior knowledge is not available, we can set the value of γ to be sufficiently large such that the condition about γ can be satisfied. We can determine whether the setting of γ is suitable by monitoring the convergence of the residual error $\|A(t)\mathbf{x}(t) - \mathbf{b}(t)\|_2$.

According to Theorems 1–3, the proposed FTCGNN method is not sensitive to initial conditions, i.e., global convergence is theoretically guaranteed given that the model parameters are properly set. Such a property is desirable for practical applications.

IV. SIMULATIONS AND COMPARISONS

In this section, simulation results of the proposed FTCGNN model are discussed. We also emphasize our FTCGNN model's advantages in terms of its ability to solve time-varying SLEs with robustness to noises by conducting comparisons with the state-of-the-art methods for solving this problem.

A. Static SLE

We first consider a static SLE given in [44], of which

$$A = \begin{bmatrix} 2.2 & 5.7 & 1 & 3.8 & 3.5 & 6.9 \\ 3.1 & 0 & 2.5 & 0 & 3.5 & 7.9 \\ 8.9 & 2.8 & 4.3 & 1 & 1.3 & 1 \\ 3 & 3.6 & 5.2 & 4.9 & 10.6 & 2.3 \\ 5.8 & 0 & 2.8 & 0 & 5.2 & 6.8 \\ 2.1 & 6.3 & 0 & 3.5 & 6.9 & 5.4 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 3.1 \\ 0 \\ 2.7 \\ 0 \\ 5.3 \\ 1.9 \end{bmatrix} \quad (20)$$

with the solution being

$$\mathbf{x}^* \approx [2.5984, -3.4402, -3.7138, 5.1895, -0.1967, 0.2428]^T.$$

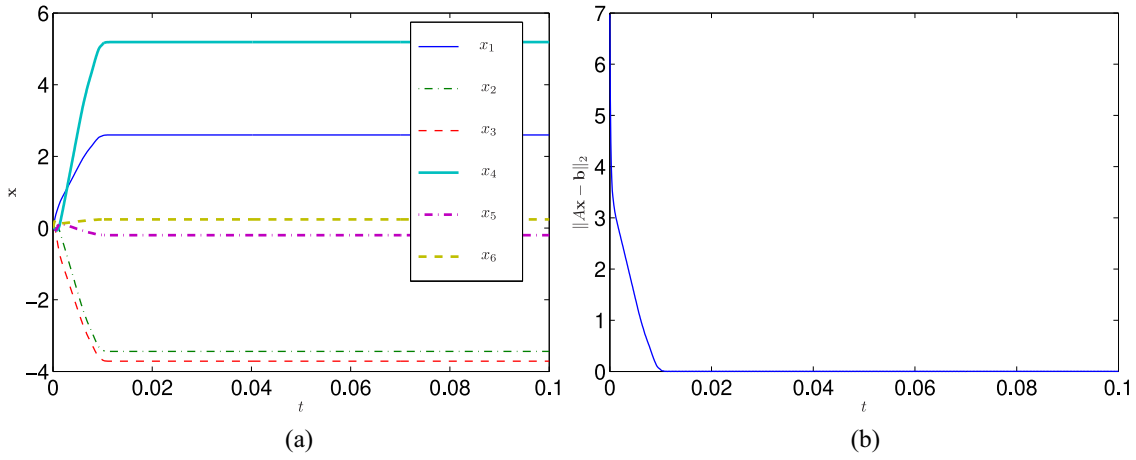


Fig. 2. State trajectories and solution error $\|Ax - b\|_2$ of the FTCGNN model (9) for solving static SLE specified by (20) with $\gamma = 1000, p = 1$, and $\mathbf{x}(0) = [0, 0, 0, 0, 0, 0]^T$. (a) State trajectories. (b) Solution error.

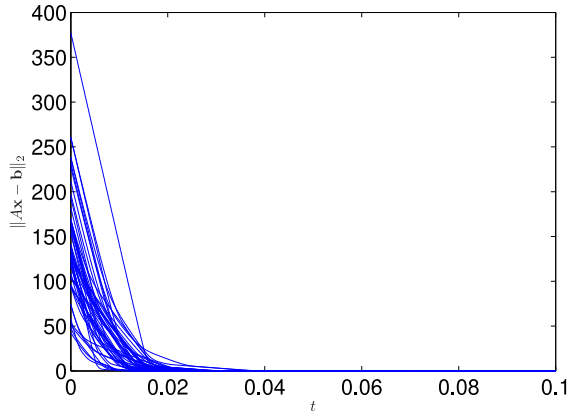


Fig. 3. Solution errors $\|Ax - b\|_2$ of the FTCGNN model (9) for solving static SLE (1) specified by (20) with $\gamma = 1000, p = 1$, and 50 sets of initial values randomly generated within the region $[-10, 10]^6$ for state variable $\mathbf{x}$.

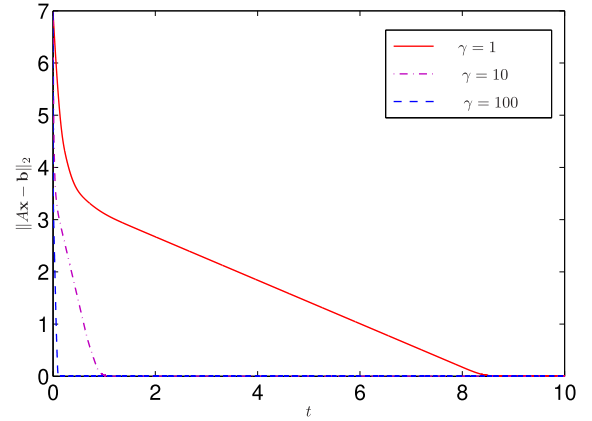


Fig. 4. Solution errors $\|Ax - b\|_2$ of the FTCGNN model (9) for solving static SLE (1) specified by (20) with $p = 1, \mathbf{x}(0) = [0, 0, 0, 0, 0, 0]^T$, and different values of γ .

With $\gamma = 1000, p = 1$ and $\mathbf{x}(0) = [0, 0, 0, 0, 0, 0]^T$, simulation is conducted for the proposed FTCGNN model (9), and the state evolution and solution error $\|Ax - b\|_2$ are depicted in Fig. 2. According to Theorem 1, the FTCGNN model (9) can solve SLE specified by (20) in finite time $t_c < 35.2621$. As seen from Fig. 2, $\mathbf{x}$ of FTCGNN model (9) converges to $\mathbf{x}^*$ of the SLE specified by (20) in finite time, i.e., less than 0.02, which is consistent with Theorem 1. To test the convergence region of the FTCGNN model (9), 50 sets of initial values randomly generated within the region $[-10, 10]^6$ for $\mathbf{x}$ are tested, and the solution errors are shown in Fig. 3. Stating from any initial values, the solution error $\|Ax - b\|_2$ synthesized by the FTCGNN model (9) converges, showing the global finite-time convergence of the model. To show the effect of the parameter γ on the convergence rate, under the same initial value with $p = 1$, simulations are conducted for the FTCGNN model (9) with $\gamma = 1, \gamma = 10$, and $\gamma = 100$. As shown in Fig. 4, larger values of γ lead to faster convergence of the solution error $\|Ax - b\|_2$, which coincides with Theorem 1.

B. Time-Varying SLE

In this section, a dynamic SLE given in [35] is considered with

$$\begin{aligned} A(t) &= \begin{bmatrix} 0.3 \cos(t) + 0.2 & 0.3 \sin(t) \\ -0.3 \sin(t) & 0.3 \cos(t) - 0.2 \end{bmatrix} \\ \mathbf{b}(t) &= \begin{bmatrix} -0.5 \cos(t) \\ 0.5 \sin(t) \end{bmatrix} \end{aligned} \quad (21)$$

and the solution is

$$\mathbf{x}^*(t) = \begin{bmatrix} 2 \cos(t) - 3 \\ 2 \sin(t) \end{bmatrix}. \quad (22)$$

With $\gamma = 100$ and $p = 1$, the performance of FTCGNN model (9) is tested for dynamic SLE (1) specified by (21) with $\mathbf{x}(0) = [1, 1]^T$. As seen from Fig. 5, the state variable $\mathbf{x}$ of the proposed FTCGNN model (9) can track $\mathbf{x}^*$ of the time-varying SLE (21) specified by (21) in finite time, which verifies Theorem 2.

The proposed FTCGNN model is also compared with other GNN models for solving time-varying SLE (21). Specifically, the GNN model (2) activated by linear function (3), sinh

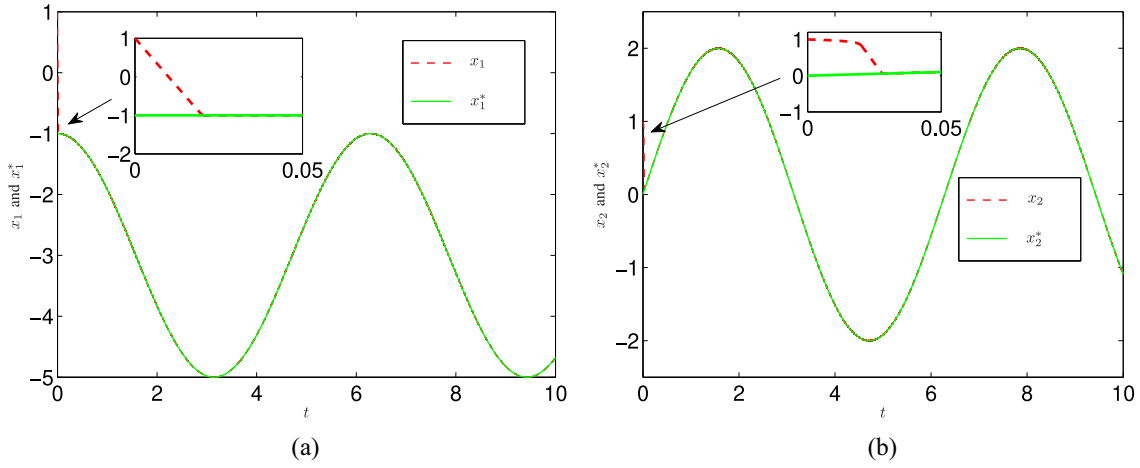


Fig. 5. Trajectories of the time-varying theoretical solution (22) denoted by $\mathbf{x}^*$ and the state variable $\mathbf{x}$ of the FTCGNN model (9) for solving time-varying SLE (1) specified by (21) with $\gamma = 100, p = 1$, and $\mathbf{x}(0) = [1, 1]^T$. (a) and (b) State trajectories.

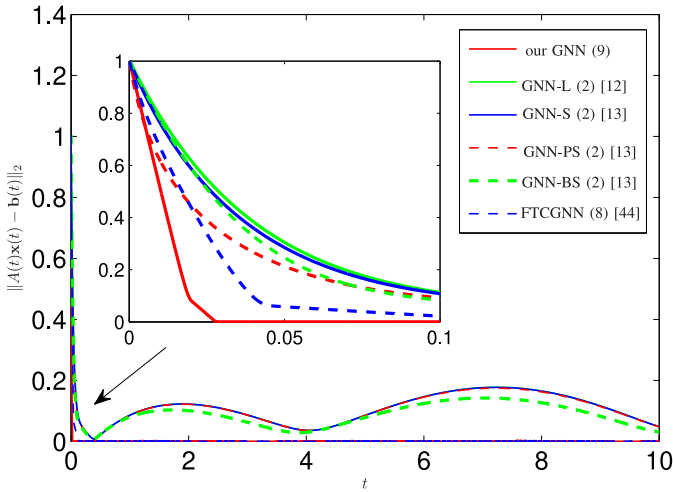


Fig. 6. Trajectories of solution error $\|A(t)\mathbf{x}(t) - \mathbf{b}(t)\|_2$ synthesized by different GNN models for solving time-varying SLE (1) specified by (21), where GNN-L, GNN-S, GNN-PS, and GNN-BS denote GNN (2) activated by linear function (3), sinh function (4), bipolar-sigmoid function (5), and powersum function (6) with $N = 5$, respectively, the initial value of $\mathbf{x}$ is set to $\mathbf{x}(0) = [1, 1]^T$, and the parameters of the GNN models are set to $\gamma = 100, p = 1, r = 0.2$.

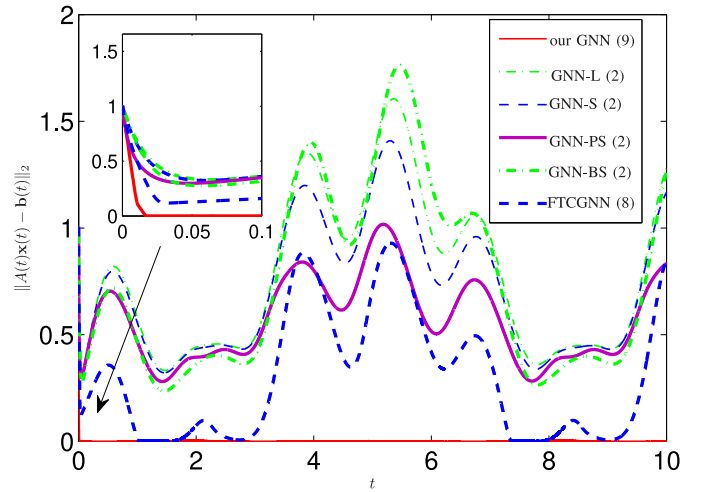


Fig. 7. Comparison of solution error $\|A(t)\mathbf{x}(t) - \mathbf{b}(t)\|_2$ generated by different GNN models for solving time-varying SLE specified by (21) under noise $\sigma(t) = 20 + 10 \sin(4t)$, where GNN-L, GNN-S, GNN-PS, and GNN-BS denote GNN (2) activated by linear function (3), sinh function (4), bipolar-sigmoid function (5), and powersum function (6) with $N = 5$, respectively, the initial value of $\mathbf{x}$ is set to $\mathbf{x}(0) = [1, 1]^T$, and the parameters of the GNN models are set to $\gamma = 200, p = 1$, and $r = 0.2$.

function (4), bipolar-sigmoid function (5), and powersum function (6) with $N = 5$, which are abbreviated as GNN-L, GNN-S, GNN-PS, GNN-BS, respectively, and the FTCGNN model (8) in [44] are considered. We set $\gamma = 100, p = 1$, and $r = 0.2$ for a fair comparison. With $\mathbf{x}(0) = [1, 1]^T$, the solution errors $\|A(t)\mathbf{x}(t) - \mathbf{b}(t)\|_2$ synthesized by different GNN models for the SLE specified by (21) are shown in Fig. 6. Evidently, GNN-L, GNN-S, GNN-PS, and GNN-BS models have large solution errors $\|A(t)\mathbf{x}(t) - \mathbf{b}(t)\|_2$ for solving the dynamic SLE, while the proposed FTCGNN (9) and the FTCGNN (8) proposed in [44] can make $\|A(t)\mathbf{x}(t) - \mathbf{b}(t)\|_2$ converge. In particular, the proposed FTCGNN (9) has faster convergence than the FTCGNN (8) proposed in [44]. The comparison demonstrates the advantages of our FTCGNN (9) in terms of convergence behavior for dynamic SLEs under without noises.

C. Robustness to Noises

In this section, we further test FTCGNN model (9) in terms of robustness to noises via simulations. In the simulations, it is found that the condition regarding the setting of γ given in Theorem 3 is conservative, and the proposed FTCGNN model (9) has much better robustness to noises than our expectation. We first compare the performance of different GNN models for solving time-varying SLE (1) specified by (21) under noise $\sigma(t) = 20 + 10 \sin(4t)$. As seen from Fig. 7, when polluted by noise $\sigma(t) = 20 + 10 \sin(4t)$, the proposed FTCGNN model (9) can still make the solution error $\|A(t)\mathbf{x}(t) - \mathbf{b}(t)\|_2$ converge to zero in finite time. Meanwhile, the solution errors $\|A(t)\mathbf{x}(t) - \mathbf{b}(t)\|_2$ synthesized by other GNN models are quite large. We also compare the proposed FTCGNN model (9) with ZNN model (15) [20], FTCZNN model (16) [34], and NTZNN model (17) [24] for dynamic

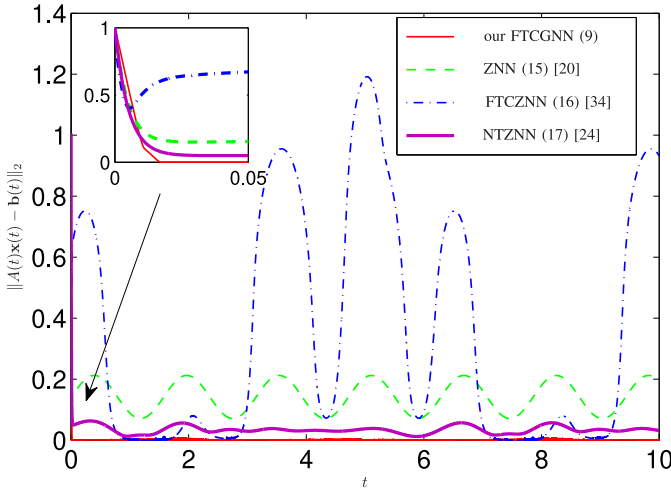


Fig. 8. Comparison of solution error $\|A(t)\mathbf{x}(t) - \mathbf{b}(t)\|_2$ generated by the proposed FTCGNN model (9), ZNN model (15) with linear activation function (3), FTCZNN model (16) with sign-bi-power activation function (7), and NTZNN model (17) for solving time-varying SLE specified by (21) under noise $\sigma(t) = 20 + 10\sin(4t)$, for which the initial value of $\mathbf{x}$ is set to $\mathbf{x}(0) = [1, 1]^T$, and the parameters of the models are set to $\gamma = 200, \kappa = 200, p = 1, r = 0.2$.

SLE (1) specified by (21) under noise $\sigma(t) = 20 + 10\sin(4t)$ under the same setting. As seen from Fig. 8, under noise $\sigma(t) = 20 + 10\sin(4t)$, the NTZNN model (17) proposed in [24] has much smaller solution error $\|A(t)\mathbf{x}(t) - \mathbf{b}(t)\|_2$ than the ZNN model (15) [20] and the FTCZNN model (16) [34], but still cannot make the solution error converge. However, via the proposed FTCGNN model (9), the solution error $\|A(t)\mathbf{x}(t) - \mathbf{b}(t)\|_2$ quickly converges to zero in finite time (i.e., less than 0.05). The above results demonstrate the advantages of the proposed FTCGNN model (9) for solving time-varying SLEs in terms of robustness to noises.

Remark 6: Our work provides a modified GNN model for solving SLEs, and it belongs to the GNN family. As a result, it makes sense to compare the proposed GNN model with existing GNN models for solving SLEs. This comparison shows that our modification does contribute to better performance for the GNN model. Meanwhile, in this article, we also compare our GNN model (9) with ZNN models for solving SLEs. It should be noted that ZNN models require knowing explicit formulas or exact values of time-derivatives of variables involved in the SLEs, and they are state-of-the-art method for solving time-varying SLEs. By simulation, we show that our GNN model (9) outperforms ZNN models in the case with noises, showing the advantages of the our GNN model compared with ZNN models in terms of robustness. Additionally, we also conduct simulation comparison with the most advanced FTCZNN model in [34], and the comparison validates the advantages of the proposed GNN model (9) in the case with noise. It should be noted that ZNN models use explicit formulas or exact values of time-derivatives of variables involved in the SLEs, but our GNN model (9) does not use such information, i.e., our model uses less information than the ZNN models. Finally, we would like to clarify that our work propose a FTCGNN model for solving SLEs, and our model does not belong to the ZNN family.

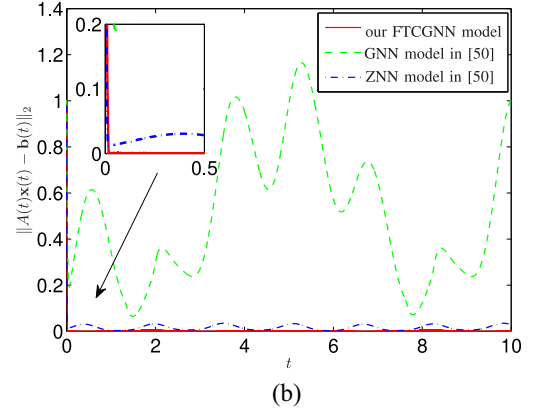
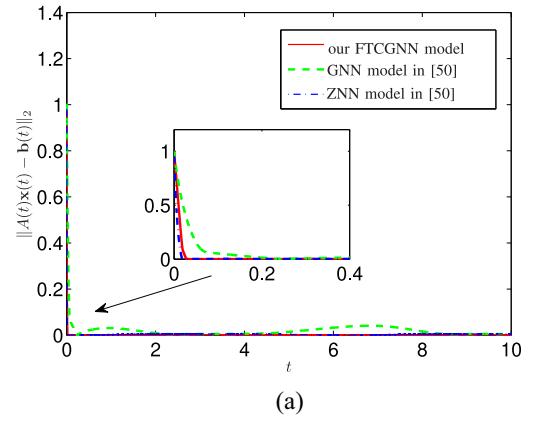


Fig. 9. Comparison of solution error $\|A(t)\mathbf{x}(t) - \mathbf{b}(t)\|_2$ generated by different models for solving time-varying SLE specified by (21), for which the initial value of $\mathbf{x}$ is set to $\mathbf{x}(0) = [1, 1]^T$, and the parameters of the models are set to $p = k_1 = k_2 = k_3 = 1, \rho = 0.2$, and $\lambda = 10^{-3}$. (a) Case with no noise and $\gamma = 100$. (b) With noise $\sigma(t) = 20 + 10\sin(4t)$ and $\gamma = 200$. (a) No noise. (b) With noise.

D. Comparison With [50]

In light of the comment of an anonymous reviewer, in this section, we would like to clarify that our model is not a special case of the models in [50].

Let

$$\text{Lip}^\rho(x_i) = \begin{cases} x_i^\rho, & x_i > 0 \\ 0, & x_i = 0 \\ -x_i^\rho, & x_i < 0 \end{cases}$$

where $\rho \in (0, 1)$. The models in [50] and our FTCGNN model are listed in Table I for clear comparison. Compared with the ZNN model in [50], our FTCGNN model does not include the terms $\lambda I, \dot{A}(t), \dot{\mathbf{b}}(t)$, or the activation function $\Phi(\cdot)$, but has an additional term $\|A^T(t)(A(t)\mathbf{x}(t) - \mathbf{b}(t))\|_2^\rho$. Compared with the GNN model in [50], our FTCGNN model does not have the terms λI or the activation function $\Phi(\cdot)$, but has an additional term $\|A^T(t)(A(t)\mathbf{x}(t) - \mathbf{b}(t))\|_2^\rho$. Thus, our FTCGNN model cannot be viewed as a special case of the ZNN model or the GNN model in [50].

We perform simulations to show the performance difference between our model and the models in [50] using the time-varying SLE (21). In the simulation, the parameters are set to $k_1 = k_2 = k_3 = p = 1, \lambda = 10^{-3}$, and $\rho = 1/3$ according to [50]. As seen from Fig. 9(a), under no noise and $\gamma = 100$,

TABLE I
COMPARISON OF THE MODELS IN [50] AND OUR FTCGNN MODEL (9) FOR SOLVING TIME-VARYING SLEs

	mathematical model
our FTCGNN model	$\ A^T(t)(A(t)\mathbf{x}(t) - \mathbf{b}(t))\ _2^p \dot{\mathbf{x}}(t) = -\gamma A^T(t)(A(t)\mathbf{x}(t) - \mathbf{b}(t))$
ZNN model in [50]	$A(t)\dot{\mathbf{x}} = \mathbf{b}(t) - A(t)\mathbf{x}(t) - \gamma\Phi((A(t) + \lambda I)\mathbf{x}(t) - \mathbf{b}(t))$
GNN model in [50]	$\dot{\mathbf{x}}(t) = -\gamma(A(t) + \lambda I)^T\Phi((A(t) + \lambda I)\mathbf{x}(t) - \mathbf{b}(t))$

Note: $\Phi(x_i) = k_1 \text{Lip}^\rho(x_i)/2 + k_2 \text{Lip}^{1/\rho}(x_i)/2 + k_3 x_i/2$, $\lambda > 0 \in \mathbb{R}$.

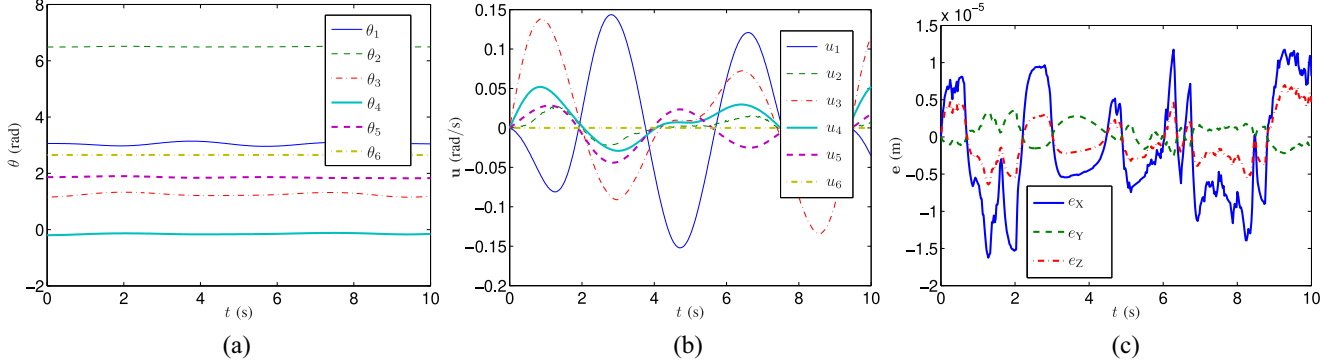


Fig. 10. Relevant data profiles for the UR5 manipulator equipped with FTCGNN model (9) with $\gamma = 10$ by solving time-varying SLE (23) with $\vartheta = 10$ and the desired trajectory given in (24), where $\mathbf{e} = [e_x, e_y, e_z]^T = \mathbf{r} - \mathbf{r}_d$. (a) Joint angle profiles. (b) Velocity command profiles. (c) Profiles of end-effector tracking error $\mathbf{e}$.

FTCGNN model (9) outperforms the GNN model in [50]. Meanwhile, in this case, the performance of our FTCGNN model is comparable with the ZNN model in [50], although our FTCGNN model (9) does not use the real-time information of $\dot{A}(t)$ and $\dot{\mathbf{b}}(t)$. As seen from Fig. 9(b), for the case with noise $\sigma(t) = 20 + 10 \sin(4t)$ and $\gamma = 200$, the proposed FTCGNN model (9) outperforms the GNN model and the ZNN model in [50]. In fact, as seen from the figure, the solution error $\|A(t)\mathbf{x}(t) - \mathbf{b}(t)\|_2$ of the other two models do not converge. These results validate the advantages of the proposed model in terms of robustness to noises and convergence speed. The simulation results also indicate that our model (9) is not a special case of the models in [50], and in fact, our model (9) is of better performance than those in [50].

E. Application to Kinematic Control of Manipulator

One potential application of the proposed model can be the kinematic control of manipulators. The kinematic control problem of a serial redundant manipulator can be described as a time-varying SLE problem as follows [49]:

$$J(\theta(t))\mathbf{u}(t) = \dot{\mathbf{r}}_d(t) - \vartheta(\mathbf{r}(t) - \mathbf{r}_d(t)) \quad (23)$$

where $J(\theta(t)) \in \mathbb{R}^{m \times n} = \partial f(\theta(t))/\partial \theta(t)$ with $n > m$ is the real-time Jacobian matrix of the corresponding forward kinematics $\mathbf{r}(t) = f(\theta(t))$ of the manipulator with $\mathbf{r} \in \mathbb{R}^m$ denoting the end-effector coordinate, $\mathbf{u} = \dot{\theta} \in \mathbb{R}^n = d\theta/dt$ is the control input with θ denoting the joint angle vector of the manipulator, and $\dot{\mathbf{r}}_d = d\mathbf{r}_d/dt$ with $\mathbf{r}_d \in \mathbb{R}^m$ being the desired end-effector trajectory. We consider a UR5 robot manipulator with $n = 6$ and $m = 3$ with $\mathbf{r}(t) = [r_x(t), r_y(t), r_z(t)]^T$, and the simulation is performed by combining MATLAB and

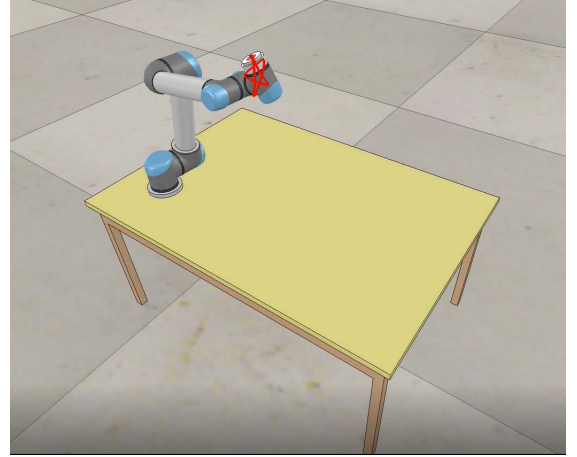


Fig. 11. End-effector trajectory of the UR5 manipulator in V-rep.

V-rep with $\vartheta = 10$. The desired trajectory is given as

$$\mathbf{r}_d(t) = \begin{bmatrix} 0.4(\cos(2t)/25 + \cos((4t)/3)/10) + r_x(0) - 0.056 \\ 0.4(\sin(2t)/25 - (3 \sin((4t)/3))/50) + r_y(0) \\ r_z(0) \end{bmatrix} \quad (24)$$

which describes a star-shape trajectory. The proposed FTCGNN model (9) is adopted with $\gamma = 10$. As seen from Fig. 10, the position error is very small (less than 2×10^{-5} m), and the generated velocity commands and joint angle profiles are smooth. As seen from Fig. 11, the end-effector trajectory of the UR5 manipulator is of the star shape which is consistent with the desired one described by (24). The results substantiate the effectiveness of the proposed FTCGNN model (9) for the kinematic control of manipulators.

V. CONCLUSION

In this article, a FCGNN model has been proposed to solve both static and time-varying SLEs. Without knowing the explicit formulas or exact values of derivative information about time-varying SLEs, under mild conditions, the state variable of the proposed GNN has finite-time convergence to the corresponding theoretical solution regardless of the existence of time-varying noises. The effectiveness and superiority of the proposed GNN model have been validated by simulation results, including comparisons with the state-of-the-art methods for time-varying problem solving. An application to the kinematic control of a UR5 manipulator has also substantiated the potential impact of the proposed model. Nonlinear control problems are widely encountered in engineering applications. There have been some works that convert nonlinear control problems into time-varying optimization problems, such as [51], [52], and [53]. As a result, it is possible to extend the current method to deal with nonlinear control problems by converting the problem into a time-varying optimization problem. It would be interesting to investigate the extension of the proposed model to deal with time-varying optimization problems for nonlinear control. Given that matrix A is uncertain, the accurate solution of SLEs cannot be found. The extension of the proposed method for the case that matrix A of SLEs is uncertain will be our future work.

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